

Algebraic and Photonic Realization of Hadamard and CNOT Gates

Matrix-Based Mapping Compared with Perceval Dual-Rail
Implementation

Saher Zayed Presented to Prof. Ahmed H. Elmahdy

June 4, 2026

1 Photonic Dual-Rail Encoding

In Perceval, a logical qubit is represented using two optical modes. The logical value is encoded by the location of a single photon:

$$|0\rangle \equiv |1, 0\rangle$$

$$|1\rangle \equiv |0, 1\rangle$$

Therefore, for a two-qubit state, four optical modes are used. The first two modes represent the control qubit and the second two modes represent the target qubit:

$$|00\rangle \equiv |1, 0, 1, 0\rangle$$

$$|01\rangle \equiv |1, 0, 0, 1\rangle$$

$$|10\rangle \equiv |0, 1, 1, 0\rangle$$

$$|11\rangle \equiv |0, 1, 0, 1\rangle$$

2 Hadamard Gate

2.1 Algebraic Mapping

Assume the input qubit is:

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

The Hadamard matrix is:

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Applying the Hadamard gate to the input state gives:

$$H|0\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Therefore:

$$|0\rangle \rightarrow \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

The measurement probabilities are:

$$P(|0\rangle) = \frac{1}{2}, \quad P(|1\rangle) = \frac{1}{2}$$

2.2 Perceval Photonic Counterpart

Using dual-rail encoding, the logical input state $|0\rangle$ is encoded as:

$$|0\rangle \equiv |1, 0\rangle$$

In Perceval, the photonic Hadamard is implemented using a 50/50 beam splitter, shown in Figure 1. The beam splitter mixes the two optical modes and produces an equal probability of finding the photon in either rail.

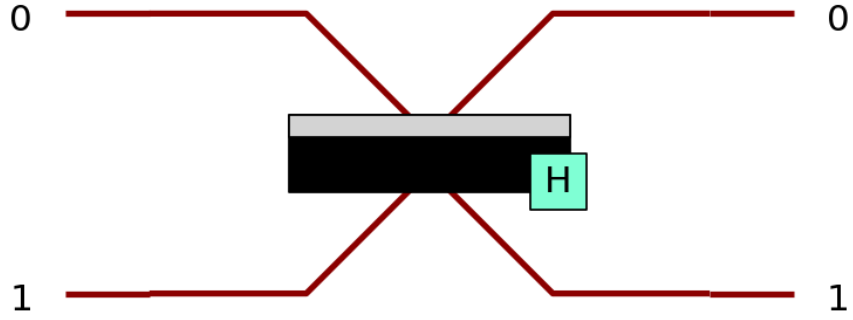


Figure 1: Perceval photonic Hadamard implementation using a 50/50 beam splitter.

The Perceval output is:

$$|1, 0\rangle : 0.5$$

$$|0, 1\rangle : 0.5$$

This matches the algebraic result:

$$|0\rangle \rightarrow \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

Thus, the Perceval beam splitter reproduces the same logical Hadamard behavior as the analytical matrix model.

3 CNOT Gate

3.1 Algebraic Mapping

The input state is:

$$|10\rangle$$

This means:

$$\text{control} = 1, \quad \text{target} = 0$$

The CNOT rule is:

if control = 0, target unchanged

if control = 1, target flips

Therefore:

$$|10\rangle \rightarrow |11\rangle$$

Using the computational basis ordering:

$$|00\rangle, |01\rangle, |10\rangle, |11\rangle$$

The CNOT matrix is:

$$\text{CNOT} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

The input vector is:

$$|10\rangle = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

Applying CNOT:

$$\text{CNOT}|10\rangle = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} = |11\rangle$$

3.2 Perceval Photonic Counterpart

In Perceval, the same logical input $|10\rangle$ is represented using dual-rail encoding as:

$$|10\rangle \equiv |0, 1, 1, 0\rangle$$

Here:

- modes 0 and 1 encode the control qubit,
- modes 2 and 3 encode the target qubit.

The Perceval implementation uses a postprocessed photonic CNOT processor, shown in Figure 2. Unlike the abstract CNOT matrix, the photonic version requires beam splitters, carefully chosen phase/reflectivity parameters, heralding, and postselection to reproduce the two-qubit logical operation.

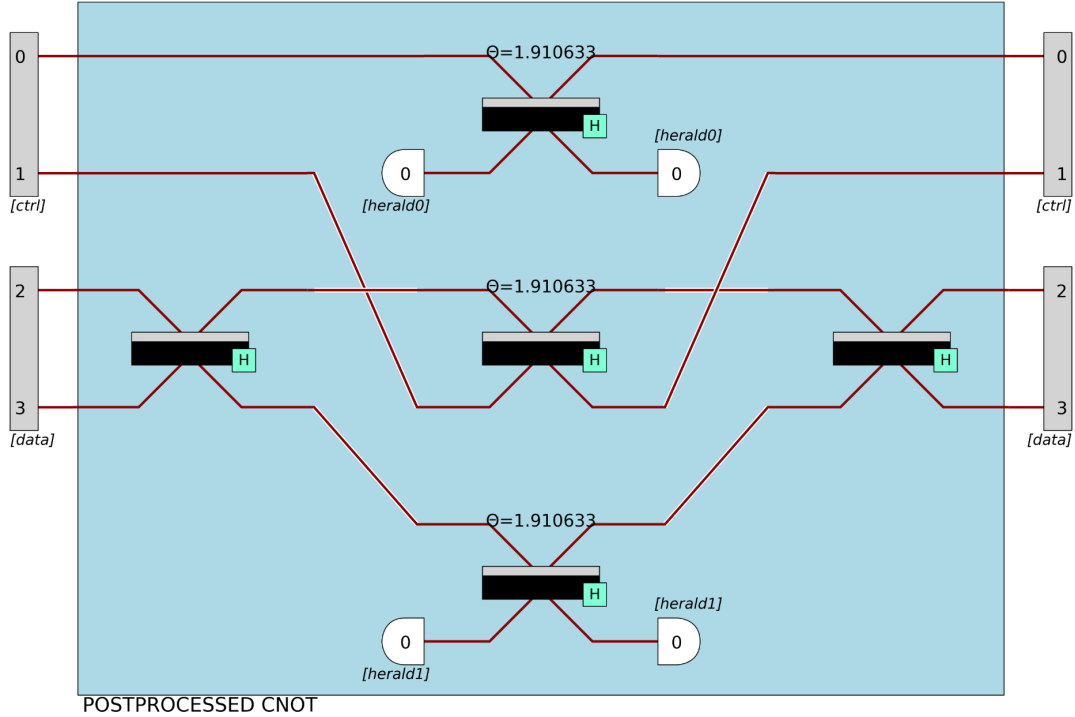


Figure 2: Perceval postprocessed photonic CNOT processor. Modes 0–1 represent the control qubit and modes 2–3 represent the target/data qubit.

For the chosen example:

$$|10\rangle \equiv |0, 1, 1, 0\rangle$$

The CNOT output was:

$$|11\rangle \equiv |0, 1, 0, 1\rangle$$

Therefore, the photonic mapping is:

$$|0, 1, 1, 0\rangle \rightarrow |0, 1, 0, 1\rangle$$

which is equivalent to the algebraic CNOT mapping:

$$|10\rangle \rightarrow |11\rangle$$

4 Complete CNOT Truth Table in Dual-Rail Form

Logical Input	Fock Input	Logical Output	Fock Output
$ 00\rangle$	$ 1, 0, 1, 0\rangle$	$ 00\rangle$	$ 1, 0, 1, 0\rangle$
$ 01\rangle$	$ 1, 0, 0, 1\rangle$	$ 01\rangle$	$ 1, 0, 0, 1\rangle$
$ 10\rangle$	$ 0, 1, 1, 0\rangle$	$ 11\rangle$	$ 0, 1, 0, 1\rangle$
$ 11\rangle$	$ 0, 1, 0, 1\rangle$	$ 10\rangle$	$ 0, 1, 1, 0\rangle$

Table 1: CNOT truth table represented using photonic dual-rail Fock states.

Conclusion

The algebraic and photonic implementations produce the same logical behavior. Algebraically, the Hadamard maps $|0\rangle$ into an equal superposition, while the Perceval beam splitter maps $|1, 0\rangle$ into equal probability outputs $|1, 0\rangle$ and $|0, 1\rangle$. Similarly, the CNOT matrix maps $|10\rangle$ to $|11\rangle$, and the Perceval photonic CNOT maps the corresponding Fock state $|0, 1, 1, 0\rangle$ to $|0, 1, 0, 1\rangle$. This confirms that the Perceval optical circuits reproduce the expected logical quantum gate behavior using physical photonic components.